

Complex Numbers and Trigonometry I

1. Compute

$$(1+i)^{-1}, (1+2i)^2, \frac{3}{1-4i}, \sqrt{i}.$$

2. Express

$$3, -2+2i, 1-i\sqrt{2}, 1+i\sqrt{2}$$

in polar form.

3. Let $z = a + bi$. Show that

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \operatorname{Im} z = \frac{z - \bar{z}}{2i}.$$

$\operatorname{Re} z$ denotes the real part of z and $\operatorname{Im} z$ the imaginary. For example, if $\gamma = \alpha + \beta i$, then $\operatorname{Re} \gamma = \alpha$, $\operatorname{Im} \gamma = \beta$.

4. For two complex numbers z_1 and z_2 , show that

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2.$$

5. For two complex numbers z_1 and z_2 , find

$$|z_1 + z_2|^2, |z_1 - z_2|^2.$$

6. (a) Show that $\operatorname{Re} z \leq |z|$ for any complex number z . (b) Using problem 5 and part (a) of this problem, show that

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

7. Using the Cartesian distance formula

$$d(p, q) = \sqrt{(p_x - q_x)^2 + (p_y - q_y)^2},$$

show that the distance between two points $v_1 = (r_1, \theta_1)$ and $v_2 = (r_2, \theta_2)$ on the polar coordinates is given by

$$d(v_1, v_2) = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2 - \theta_1)}.$$

This result is also known as the law of cosines.

8. Let $z = x + iy$ and $\gamma = a + bi$. Let $r > 0$ be a real number. Express

$$|z - a| = r$$

only in terms of x, y, a, b , and r . Which geometric figure does this equation represent?

9. Let v be the unit vector $(1, 0)$. Let α and β be two nonzero (or non- $2\pi n$) angles. Rotate v by α and $\alpha + \beta$ to obtain v' and v'' , respectively. **(a)** Drop a perpendicular from v' to the positive x -axis. Label the resulting right triangle T_1 . What are its side lengths? **(b)** Drop a perpendicular from v'' to the segment from the origin to v' . Label the resulting right triangle T_2 . What are its side lengths? **(c)** Finally, drop a perpendicular from v'' to the positive x -axis to form T_3 . Using this construction, show that

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha,$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

10. Using the results from 9, show that for two complex numbers $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$,

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2.$$

What does this result indicate geometrically?